

Lecture 6

Gradient Descent, Continued; Foundations of Probability; Combinatorics

DSC 40A, Summer 2026

Lecture 6 Part 1

Gradient Descent, Continued

DSC 40A, Summer 2026

Announcements before break

Agenda

- Recap: Gradient descent.
- Convexity.
- More examples.
 - Huber loss.
 - Gradient descent with multiple variables.

Question 🤔

Answer in chat

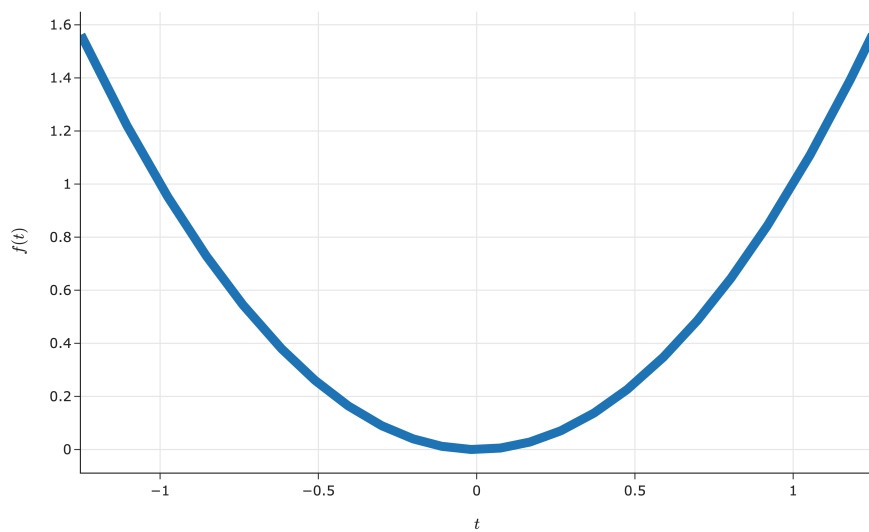
Remember, you can always ask questions in Q&A!

Overview: Gradient descent

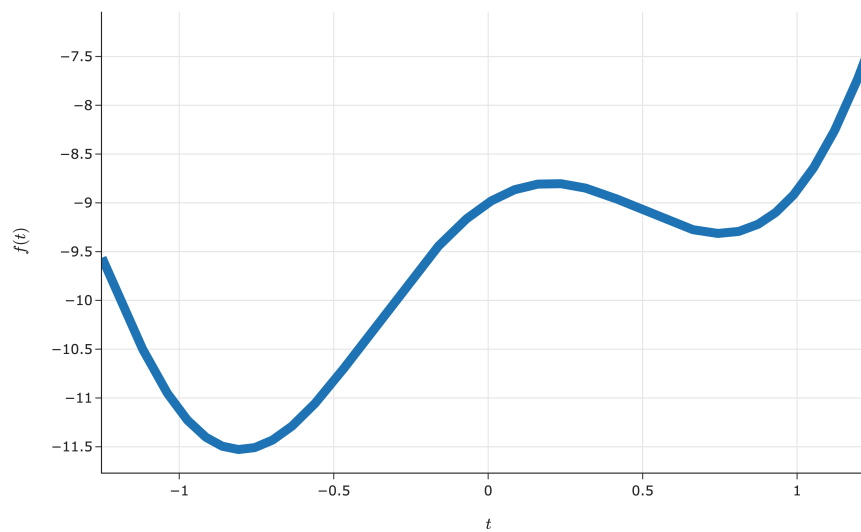
When is gradient descent guaranteed to work?

Convex functions

A convex function ✓



A non-convex function ✗

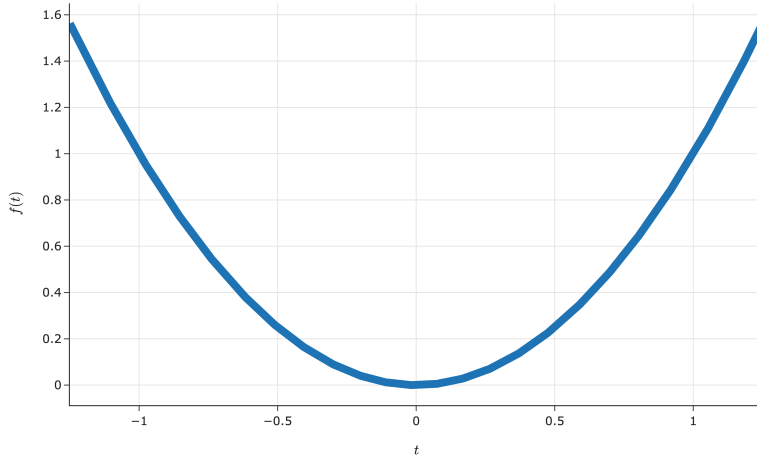


Convexity

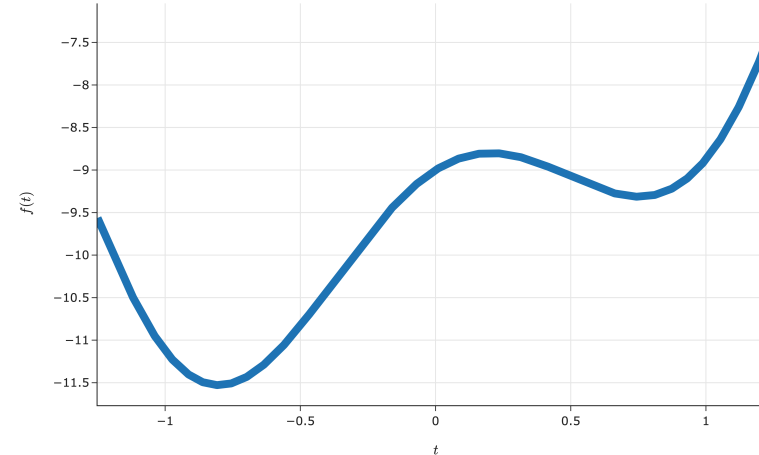
- A function f is **convex** if, for **every** a, b in the domain of f , the line segment between:
 $(a, f(a))$ and $(b, f(b))$

does not go below the plot of f .

A **convex** function ✓



A **non-convex** function ✗

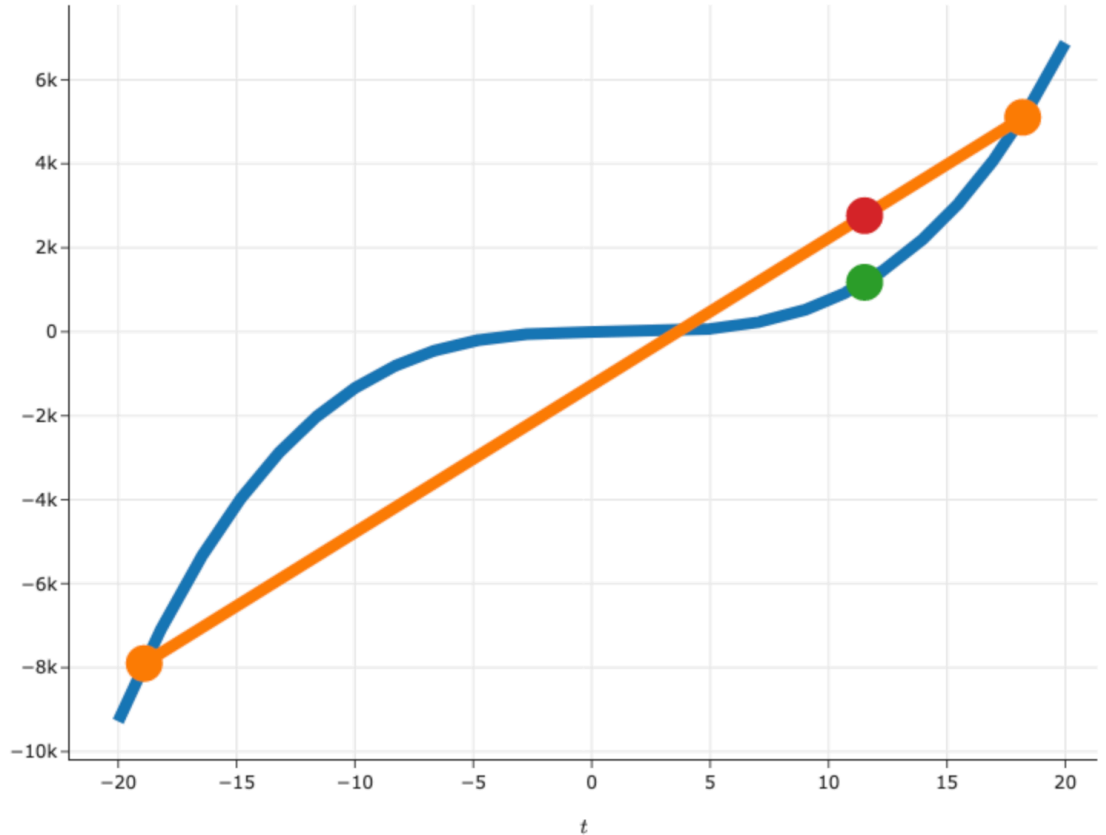


Formal definition of convexity

- A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is **convex** if, for every a, b in the domain of f , and for every $t \in [0, 1]$:

$$(1 - t)f(a) + tf(b) \geq f((1 - t)a + tb)$$

- This is a formal way of restating the definition from the previous slide.



Question 🤔

Answer in chat

Which of these functions are **not** convex?

- A. $f(x) = |x|$.
- B. $f(x) = e^x$.
- C. $f(x) = \sqrt{x - 1}$.
- D. $f(x) = (x - 3)^{24}$.
- E. More than one of the above are non-convex.

Second derivative test for convexity

- If $f(t)$ is a function of a single variable and is **twice** differentiable, then $f(t)$ is convex if and only if:

$$\frac{d^2 f}{dt^2}(t) \geq 0, \quad \forall t$$

- Example: $f(x) = x^4$ is convex.

Why does convexity matter?

- Convex functions are (relatively) easy to minimize with gradient descent.
- **Theorem:** If $f(t)$ is convex and differentiable, then gradient descent converges to a **global minimum** of f , as long as the step size is small enough.
- **Why?**
 - Gradient descent converges when the derivative is 0.
 - For convex functions, the derivative is 0 only at one place – the global minimum.
 - In other words, if f is convex, gradient descent won't get "stuck" and terminate in places that aren't global minimums (local minimums, saddle points, etc.).

Nonconvex functions and gradient descent

- We say a function is **nonconvex** if it does not meet the criteria for convexity.
- Nonconvex functions are (relatively) difficult to minimize.
- Gradient descent **might** still work, but it's not guaranteed to find a global minimum.
 - We saw this at the start of the lecture, when trying to minimize
$$f(t) = 5t^4 - t^3 - 5t^2 + 2t - 9.$$

Choosing a step size in practice

- In practice, choosing a step size involves a lot of trial-and-error.
- In this class, we've only touched on "constant" step sizes, i.e. where α is a constant.

$$t_{i+1} = t_i - \alpha \frac{df}{dt}(t_i)$$

- **Remember:** α is the "step size", but the amount that our guess for t changes is $\alpha \frac{df}{dt}(t_i)$, not just α .
- In future courses, you'll learn about "decaying" step sizes, where the value of α decreases as the number of iterations increases.
 - Intuition: take much bigger steps at the start, and smaller steps as you progress, as you're likely getting closer to the minimum.

More examples

Example: Huber loss and the constant model

- First, we learned about squared loss,

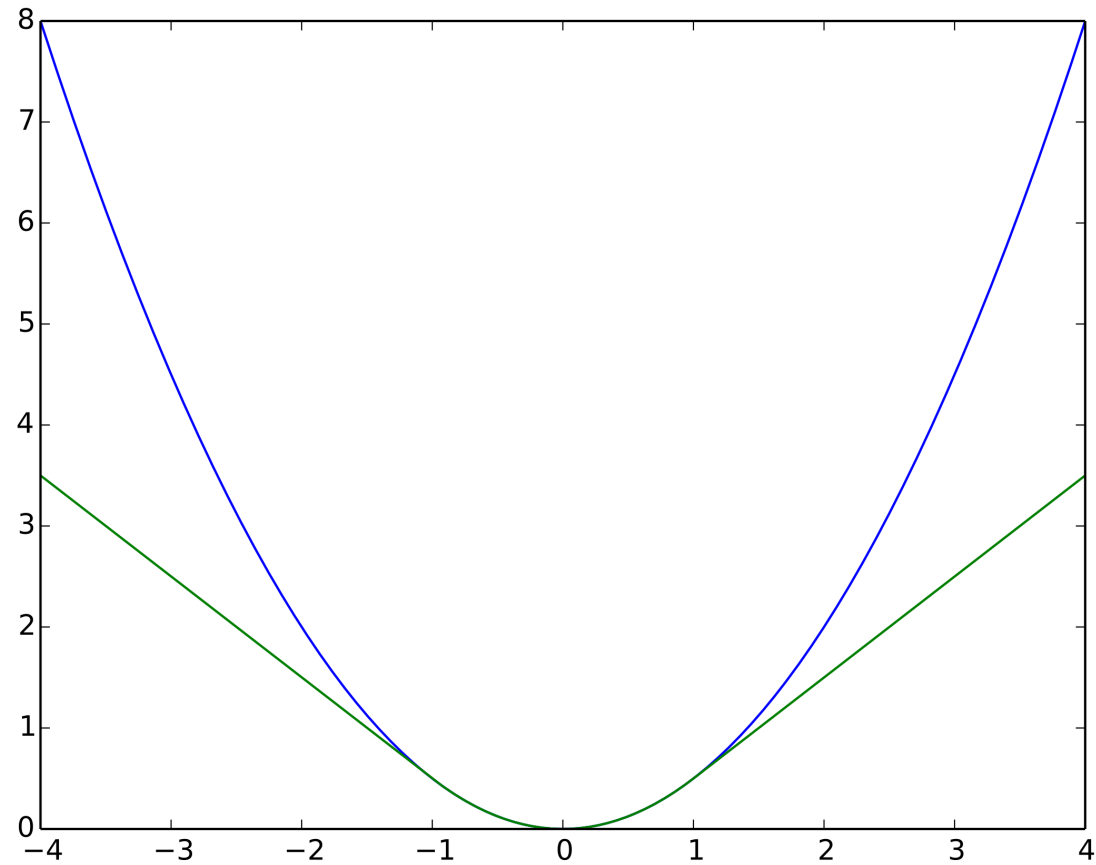
$$L_{\text{sq}}(y_i, H(x_i)) = (y_i - H(x_i))^2.$$

- Then, we learned about absolute loss,

$$L_{\text{abs}}(y_i, H(x_i)) = |y_i - H(x_i)|.$$

- Let's look at a new loss function, **Huber loss**:

$$L_{\text{huber}}(y_i, H(x_i)) = \begin{cases} \frac{1}{2} (y_i - H(x_i))^2 & \text{if } |y_i - H(x_i)| \leq \delta \\ \delta \cdot (|y_i - H(x_i)| - \frac{1}{2} \delta) & \text{otherwise} \end{cases}$$



Squared loss in blue, **Huber** loss in green.
Note that both loss functions are convex!

Minimizing average Huber loss for the constant model

- For the constant model, $H(x) = h$:

$$L_{\text{huber}}(y_i, h) = \begin{cases} \frac{1}{2}(y_i - h)^2 & \text{if } |y_i - h| \leq \delta \\ \delta \cdot (|y_i - h| - \frac{1}{2}\delta) & \text{otherwise} \end{cases}$$

$$\implies \frac{\partial L}{\partial h}(h) = \begin{cases} -(y_i - h) & \text{if } |y_i - h| \leq \delta \\ -\delta \cdot \text{sign}(y_i - h) & \text{otherwise} \end{cases}$$

- So, the **derivative** of empirical risk is:

$$\frac{dR_{\text{huber}}}{dh}(h) = \frac{1}{n} \sum_{i=1}^n \begin{cases} -(y_i - h) & \text{if } |y_i - h| \leq \delta \\ -\delta \cdot \text{sign}(y_i - h) & \text{otherwise} \end{cases}$$

- It's **impossible** to set $\frac{dR_{\text{huber}}}{dh}(h) = 0$ and solve by hand: we need gradient descent!

Lecture 6 Part 2

Foundations of Probability

DSC 40A, Summer 2026

Announcements

- Reminder! Community OH here after lecture!
- Homework 5 is due on **Tuesday at 11:59PM**.
- Congrats on finishing the midterm!
 - The final is coming up (very quick turnaround) on Friday, July 31st.
 - There are no plans to offer any alternate times since this was posted during enrollment, prior to the start of the term.

Agenda

- Overview: Probability and statistics.
- Complement, addition, and multiplication rules.
- Conditional probability.

Note: There are no more DSC 40A-specific readings, but we've posted **many** probability resources on the resources tab of the course website. These will come in handy! Specific resources you should look at:

- The DSC 40A probability roadmap, written by Janine Tiefenbruck.
- The textbook Theory Meets Data, which explains many of the same ideas and contains more practice problems.

Question 🤔

Answer in chat

Remember, you can always ask questions in Q&A!

Overview: Probability and statistics

From Lecture 1: Course overview

Part 1: Learning from Data

- Summary statistics and loss functions; empirical risk minimization.
- Linear regression (including multiple variables); linear algebra.

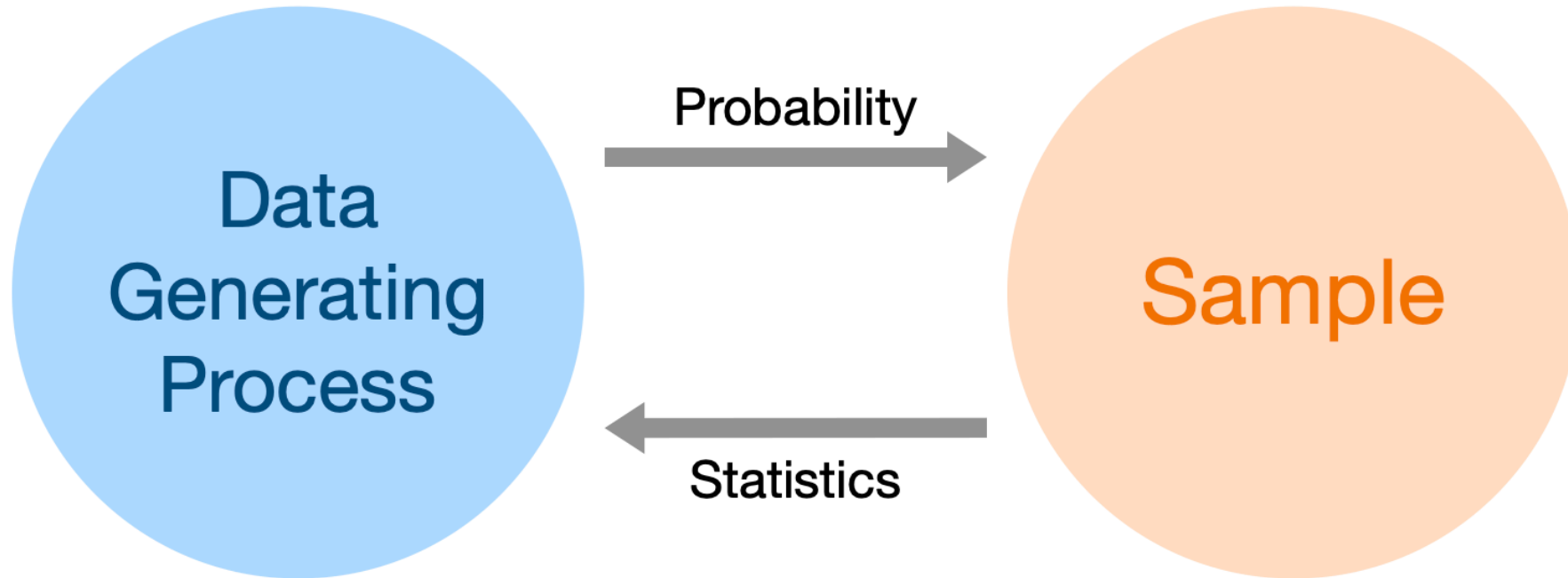
Part 2: Probability

- Set theory and combinatorics; probability fundamentals.
- Conditional probability and independence.
- The Naïve Bayes classifier.

Why do we need probability?

- So far in this class, we have made predictions based on a dataset.
- This dataset can be thought of as a **sample** of some **population**.
- For a hypothesis function to be useful in the future, the sample that was used to create the hypothesis function needs to look similar to samples that we'll see in the future.

Probability and statistics



The plan

- Lecture 6 (today): Key rules of probability. Start combinatorics.
- Lectures 7: More combinatorics. Bayes theorem.
- Lectures 8: Conditional independence and the Naïve Bayes classifier.
- Finish up probability topics. Review.

Terminology

- An **experiment** is some process whose outcome is random (e.g. flipping a coin, rolling a die).
- A **set** is an unordered collection of items.
 - Sets are usually denoted with { curly brackets }.
 - $|A|$ denotes the number of elements in set A .
- A **sample space**, S , is the set of all possible outcomes of an experiment.
 - Could be finite or infinite!
- An **event** is a subset of the sample space, or a set of outcomes.
 - $E \subseteq S$ means " E is a subset of S ."

Probability distributions

- A probability distribution, p , describes the **probability** of each outcome s in a sample space S .

- The probability of each outcome must be between 0 and 1:

$$0 \leq p(s) \leq 1$$

- The sum of the probabilities of each outcome must be exactly 1:

$$\sum_{s \in S} p(s) = 1$$

- The probability of an **event** is the sum of the probabilities of the outcomes in the event.

$$\mathbb{P}(E) = \sum_{s \in E} p(s)$$

What do probabilities *mean*?

- One interpretation: if $\mathbb{P}(E) = p$, then if we repeat our experiment infinitely many times, the proportion of repetitions in which event E occurs is p .
 - If p is large, event E occurs very frequently.
- Another interpretation: $\mathbb{P}(E) = p$ represents our "degree of belief" in the event E .
 - If p is large, we are pretty sure event E is going to happen when we perform our experiment.

Example: Probability of rolling an even number on a six-sided die

Equally-likely outcomes

- If S is a sample space with n possible outcomes, and all outcomes are equally-likely, then the probability of any one outcome occurring is $\frac{1}{n}$.
- The probability of an event E , then, is:

$$\mathbb{P}(E) = \underbrace{\frac{1}{n} + \frac{1}{n} + \dots + \frac{1}{n}}_{|E| \text{ times}} = \frac{\# \text{ of outcomes in } E}{\# \text{ of outcomes in } S} = \frac{|E|}{|S|}$$

- **Example:** Flipping a coin three times.

Complement, addition, and multiplication rules

Complement rule

- Let A be an event with probability $\mathbb{P}(A)$.
- Then, the event $\bar{A}$ is the **complement** of the event A . It contains the set of all outcomes in the sample space that are **not** in A .

$$\mathbb{P}(\bar{A}) = 1 - \mathbb{P}(A)$$

Addition rule

- We say two events are **mutually exclusive** if they have no overlap (i.e. they can't both happen at the same time).

- If A and B are mutually exclusive, then the probability that A or B happens is:

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$$

Principle of inclusion-exclusion

- If events A and B are not mutually exclusive, then the addition rule becomes more complicated.

- In general, if A and B are any two events, then:

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

Question 🤔

Answer in chat

Each day when you get home from school, there is a:

- 0.3 chance your mom is at home.
- 0.4 chance your brother is at home.
- 0.25 chance that both your mom and brother are at home.

When you get home from school today, what is the chance that **neither** your mom nor your brother are at home?

A. 0.3 B. 0.45 C. 0.55 D. 0.7 E. 0.75

Multiplication rule and independence

- The probability that events A and B both happen is

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B|A)$$

- $\mathbb{P}(B|A)$ means "the probability that B happens, given that A happened." It is a **conditional probability**.
 - More on this soon!
- If $\mathbb{P}(B|A) = \mathbb{P}(B)$, we say A and B are **independent**.
 - Intuitively, A and B are independent if knowing that A happened gives you no additional information about event B , and vice versa.
 - For two independent events, $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.

Example: Rolling a die

Let's consider rolling a fair 6-sided die. The results of each die roll are independent from one another.

- Suppose we roll the die once. What is the probability of seeing both 1 and 2?
- Suppose we roll the die once. What is the probability of seeing 1 or 2?

Example: Rolling a die

- Suppose we roll the die 3 times. What is the probability of never seeing a 1 in any of the rolls?
- Suppose we roll the die 3 times. What is the probability of seeing a 1 at least once?

Example: rolling a die

- Suppose we roll the die n times. What is the probability of only seeing the numbers 1, 3, and 4?
- Suppose we roll the die 2 times. What is the probability that the two rolls are different?

Conditional probability

Conditional probability

- The probability of an event may **change** if we have additional information about outcomes.
- Starting with the multiplication rule, $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B|A)$, we have that:

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)}$$

assuming that $\mathbb{P}(A) > 0$.

Question 🤔

Answer in chat

Suppose a family has two pets. Assume that it is equally likely that each pet is a dog or a cat. **Consider the following two probabilities:**

- The probability that both pets are dogs given that **the oldest is a dog**.
- The probability that both pets are dogs given that **at least one of them is a dog**.

Are these two probabilities equal?

- A. Yes, they're equal.
- B. No, they're not equal.

Example: Pets

Let's compute the probability that both pets are dogs given that **the oldest is a dog**.

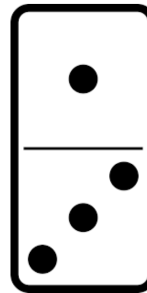
Example: Pets

Let's now compute the probability that both pets are dogs given that **at least one of them is a dog**.

Example: Dominoes

([source: 538](#))

In a set of dominoes, each tile has two sides with a number of dots on each side: 0, 1, 2, 3, 4, 5, or 6. There are 28 total tiles, with each number of dots appearing alongside each other number (including itself) on a single tile.



Example: Dominoes

Question 1: What is the probability of drawing a "double" from a set of dominoes – that is, a tile with the same number on both sides?

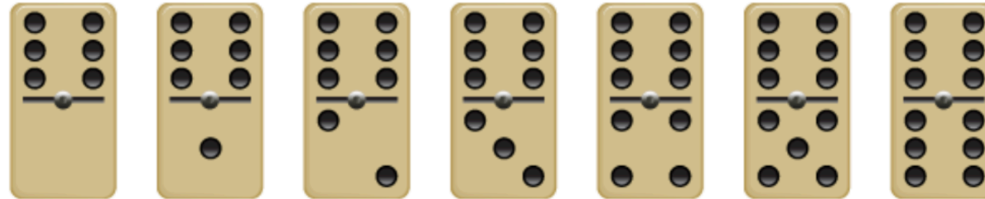
Example: Dominoes

Question 2: Now your friend picks a random tile from the set and tells you that at least one of the sides is a 6. What is the probability that your friend's tile is a double, with 6 on both sides?



Example: Dominoes

Question 3: Now you pick a random tile from the set and uncover only one side, revealing that it has 6 dots. What is the probability that this tile is a double, with 6 on both sides?



[See 538's explanation here.](#)

Summary

- If A is an event, then the complement of A , denoted $\bar{A}$, is the event that A does not happen, and $\boxed{P(\bar{A}) = 1 - P(A)}$.
- Two events A and B are mutually exclusive if they share no outcomes (no overlap). In this case, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.
- More generally, for any two events, $\boxed{\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)}$.
- The probability that events A and B both happen is $\boxed{\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B|A)}$.
- $\mathbb{P}(B|A)$ is the **conditional probability** of B occurring, given that A occurs:

$$\boxed{\mathbb{P}(B|A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)}}$$

- If $\mathbb{P}(B|A) = \mathbb{P}(B)$, then events A and B are **independent**.

Lecture 6 Part 3

Combinatorics

DSC 40A, Summer 2026

Agenda

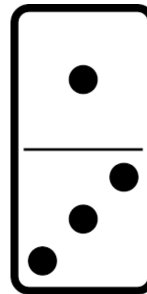
- Combinatorics.
- Lots and lots of examples.

Recap: The domino problem

Example: Dominoes

([source: 538](#))

In a set of dominoes, each tile has two sides with a number of dots on each side: 0, 1, 2, 3, 4, 5, or 6. There are 28 total tiles, with each number of dots appearing alongside each other number (including itself) on a single tile.



Combinatorics

Motivation

- Many problems in probability involve counting.
 - Suppose I flip a fair coin 100 times. What's the probability I see 34 heads?
 - Suppose I draw 3 cards from a 52 card deck. What's the probability they all are all from the same suit?
- In order to solve such problems, we first need to learn how to count.
- The area of math that deals with counting is called **combinatorics**.

Selecting elements (i.e. sampling)

- Many experiments involve choosing k elements randomly from a group of n possible elements. This group is called a **population**.
 - If drawing cards from a deck, the population is the deck of all cards.
 - If selecting people from DSC 40A, the population is everyone in DSC 40A.
- Two decisions:
 - Do we select elements with or without **replacement**?
 - Does the **order** in which things are selected matter?

Sequences

- A **sequence** of length k is obtained by selecting k elements from a group of n possible elements **with replacement (i.e. repetition is allowed)**, such that **order matters**.
- **Example:** Draw a card (from a standard 52-card deck), put it back in the deck, and repeat 4 times. How many such sequences are there?
- **Example:** A UCSD PID starts with "A" then has 8 digits. How many UCSD PIDs are possible?

Sequences

In general, the number of ways to select k elements from a group of n possible elements **with replacement (i.e. repetition is allowed) and order matters** is n^k .

Permutations

- A **permutation** is obtained by selecting k elements from a group of n possible elements **without replacement (i.e. repetition is not allowed)**, such that **order matters**.
- **Example:** Draw 4 cards, without replacement, from a standard 52-card deck. How many such permutations are there?
- **Example:** How many ways are there to select a president, vice president, and secretary from a group of 8 people?

Permutations

- In general, the number of ways to select k elements from a group of n possible elements **without replacement (i.e. repetition is not allowed)** and **order matters** is:

$$P(n, k) = (n)(n - 1) \dots (n - k + 1)$$

- To simplify: recall that the definition of $n!$ is:
- Given this, we can write:

Question 🤔

Answer in chat

UCSD has 8 colleges. In how many ways can I rank my top 3 choices?

- A. 24.
- B. 336.
- C. 512.
- D. 6561.
- E. None of the above.

Special case of permutations

- Suppose we have n people. The total number of ways I can rearrange these n people in a line is:

- This is consistent with the formula:

$$P(n, n) = \frac{n!}{(n - n)!} = \frac{n!}{0!} = \frac{n!}{1} = n!$$

- **Followup:** How many ways are there to arrange n people in a circle?

Combinations

- A **combination** is a set of k elements selected from a group of n possible elements **without replacement (i.e. repetition is not allowed)**, such that **order does not matter**.
- **Example:** There are 4 ice cream flavors. In how many ways can you pick two different flavors?

From permutations to combinations

- There is a close connection between:
 - the number of **permutations** of k elements selected from a group of n , and
 - the number of **combinations** of k elements selected from a group of n .

$$\# \text{ combinations} = \frac{\# \text{ permutations}}{\# \text{ orderings of } k \text{ elements}}$$

- Since $\# \text{ permutations} = \frac{n!}{(n-k)!}$ and $\# \text{ orderings of } k \text{ elements} = k!$, we have:

$$C(n, k) = \binom{n}{k} = \frac{n!}{(n-k)!k!}$$

Combinations

In general, the number of ways to select k elements from a group of n elements **without replacement (i.e. repetition is not allowed) and **order does not matter** is:

$$\boxed{\binom{n}{k} = \frac{n!}{(n-k)!k!}}$$

The symbol $\binom{n}{k}$ is pronounced " n choose k ", and is also known as the **binomial coefficient**.

Example: Committees

- How many ways are there to select a president, vice president, and secretary from a group of 8 people?
- How many ways are there to select a committee of 3 people from a group of 8 people?
- If you're ever confused about the difference between permutations and combinations, **come back to this example.**
- More generally, don't jump straight to a formula: think about what the question is asking for.

Aside: Simplifying $\binom{n}{k}$

It's true that:

$$\binom{n}{k} = \frac{n!}{(n-k)!k!}$$

However, when asked to simplify the value of $\binom{n}{k}$, do so strategically!

Question 🤔

Answer in chat

A domino consists of two faces, each with anywhere between 0 and 6 dots. A set of dominoes consists of every possible combination of dots on each face.

How many dominoes are in a set of dominoes?

- A. $\binom{7}{2}$
- B. $\binom{7}{1} + \binom{7}{2}$
- C. $P(7, 2)$
- D. $\frac{P(7,2)}{P(7,1)} 7!$

Summary

Suppose we want to select k elements from a group of n possible elements. The following table summarizes whether the problem involves sequences, permutations, or combinations, along with the number of relevant orderings.

	Yes, order matters	No, order doesn't matter
With replacement Repetition allowed	n^k possible sequences	more complicated: watch this video *
Without replacement Repetition not allowed	$\frac{n!}{(n-k)!}$ possible permutations	$\binom{n}{k}$ combinations

**or see the previous slide.*

More examples

All we're going to do for the remainder of today's lecture and much of Wednesday's lecture is work through examples of combinatorics problems.

Combinatorics as a tool for probability

- If S is a sample space consisting of equally-likely outcomes, and A is an event, then $\mathbb{P}(A) = \frac{|A|}{|S|}$.
- In many examples, this will boil down to using permutations and/or combinations to count $|A|$ and $|S|$.
- **Tip:** Before starting a probability problem, always think about what the sample space S is!

Overview: Selecting students

We're going to answer the same question using several different techniques.

There are 20 students in a class. Avi is one of them. Suppose we select 5 students in the class uniformly at random without replacement. What is the probability that Avi is among the 5 selected students?

Selecting students

Method 1: Using permutations

There are 20 students in a class. Avi is one of them. Suppose we select 5 students in the class uniformly at random **without replacement**. What is the probability that Avi is among the 5 selected students?

Selecting students

Method 2: Using permutations and the complement

There are 20 students in a class. Avi is one of them. Suppose we select 5 students in the class uniformly at random **without replacement**. What is the probability that Avi is among the 5 selected students?

Selecting students

Method 3: Using combinations

There are 20 students in a class. Avi is one of them. Suppose we select 5 students in the class uniformly at random **without replacement**. What is the probability that Avi is among the 5 selected students?

Selecting students

Method 4: The "easy" way

There are 20 students in a class. Avi is one of them. Suppose we select 5 students in the class uniformly at random **without replacement**. What is the probability that Avi is among the 5 selected students?

Question 🤔

Answer in chat

With vs. without replacement

We've determined that a probability that a random sample of 5 students from a class of 20 **without replacement** contains Avi (one student in particular) is $\frac{1}{4}$.

Suppose we instead sampled **with replacement**. Would the resulting probability be equal to, greater than, or less than $\frac{1}{4}$?

- A. Equal to.
- B. Greater than.
- C. Less than.

Even more examples (if time)

Watch the video linked below! (if no more time)

The following slides **will not** be covered in lecture. Instead, they're covered in [this walkthrough video](#).

1. 2 dogs and 2 cats?
2. 3 dogs and 1 cat?
3. At least 2 dogs?

① $\binom{5}{2} \cdot \binom{7}{2}$
② $\binom{5}{3} \binom{7}{1}$

③ at least 2 dogs
= 2 dogs, 2 cats OR
3 dogs, 1 cat OR
4 dogs, 0 cats
 $\Rightarrow \binom{5}{2} \binom{7}{2} + \binom{5}{3} \binom{7}{1} + \binom{5}{4} \binom{7}{0}$

A screenshot from the walkthrough video.

We'll still cover more examples on Tuesday, but now you have a few more examples to refer to when working on Homework 6 over the weekend.

Example: Pets

Part 1: We have 12 pets: 5 dogs and 7 cats. In how many ways can we select 4 pets?

Example: Pets

Part 2: We have 12 pets: 5 dogs and 7 cats. In how many ways can we select 4 pets such that we have...

1. 2 dogs and 2 cats?
2. 3 dogs and 1 cat?
3. At least 2 dogs?

Example: Pets

Part 3: We have 12 pets: 5 dogs and 7 cats. We randomly select 4 pets. What's the probability that we selected at least 2 dogs?

Example: A fair coin

Suppose we flip a fair coin 10 times.

1. What is the probability that we see the specific sequence THTTHTHHTH?
2. What is the probability that we see an equal number of heads and tails?

Example: An unfair coin

Suppose we flip an **unfair** coin 10 times. The coin is biased such that for each flip, $\mathbb{P}(\text{heads}) = \frac{1}{3}$.

1. What is the probability that we see the specific sequence THTTHTHHTH?
2. What is the probability that we see an equal number of heads and tails?

